MetaFormer Is Actually What You Need for Vision

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Code: https://github.com/sail-sg/poolformer

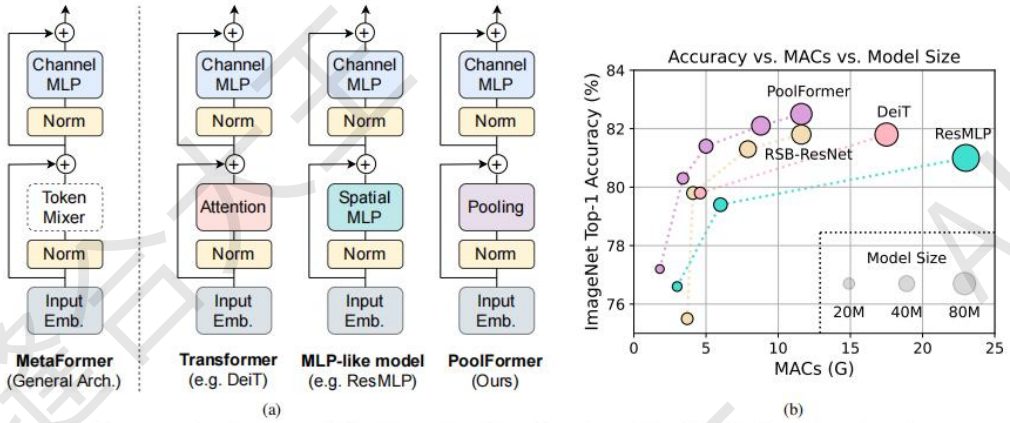

Figure 1. **MetaFormer and performance of MetaFormer-based models on ImageNet-1K validation set.** As shown in (a), we present *MetaFormer* as a general architecture abstracted from Transformers [56] by not specifying the token mixer. When using attention/spatial MLP as the token mixer, MetaFormer is instantiated as Transformer/MLP-like models. We argue that the competence of Transformer/MLP-like models primarily stems from the general architecture MetaFormer instead of the equipped specific token mixers. To demonstrate this, we exploit an embarrassingly simple non-parametric operator, *pooling*, to conduct extremely basic token mixing. Surprisingly, the resulted model *PoolFormer* consistently outperforms the well-tuned vision Transformer [17] baseline (DeiT [53]) and MLP-like [51] baseline (ResMLP [52]) as shown in (b), which well supports that MetaFormer is actually what we need to achieve competitive performance. RSB-ResNet in (b) means the results are from "ResNet Strikes Back" [59] where ResNet [24] are trained with improved training procedure for 300 epochs.

MetaFormer CVPR2022 的一篇文章。

Transformer 在计算机视觉任务中展现出了巨大潜力，人们普遍认为，其基于注意力的 token mixer 模块对其贡献最大。然而，近期研究表明，Transformer 中的注意力机制模块可以被空间 MLP 替代，且所得模型仍表现良好。

基于这一观察，论文假设 Transformer 的总体架构而非特定的 token mixer 模块，对模型性能更为重要。

为了验证这一点，论文故意用极其简单的空间池化算子替换 Transformer 中的注意力机制。令人惊讶的是，论文发现所得模型（称为 PoolFormer）在多个计算机视觉任务上取得了具有竞争力的性能。

PoolFormer 的有效性验证了论文的假设，并促使论文提出了

“MetaFormer”的概念，即从 Transformer 中抽象出来的一种不特定于 token mixer 的通用架构。

基于大量实验，论文认为 MetaFormer 是近期 Transformer 和 MLP 类模型在计算机视觉任务上取得卓越结果的关键因素。

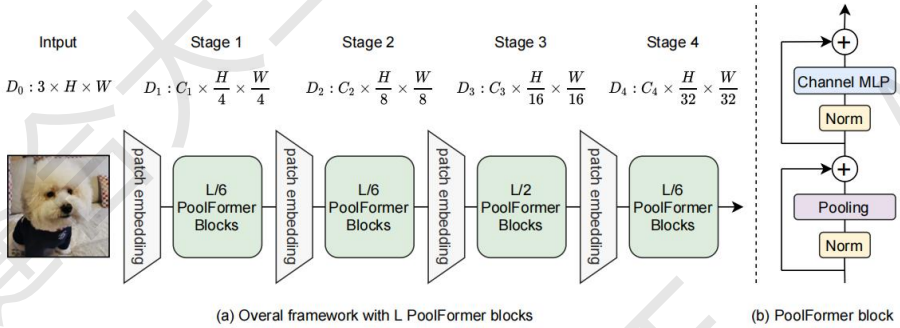

Figure 2. **(a) The overall framework of PoolFormer.** Similar to [24, 36, 57], PoolFormer adopts hierarchical architecture with 4 stages. For a model with L PoolFormer blocks, stage [1, 2, 3, 4] have [L/6, L/6, L/2, L/6] blocks, respectively. The feature dimension D_i of stage i is shown in the figure. **(b) The architecture of PoolFormer block.** Compared with Transformer block, it replaces attention with extremely simple non-parametric operator, pooling, to conduct only basic token mixing.

```
class Pooling(nn.Module):
    """
    Implementation of pooling for PoolFormer
    --pool_size: pooling size
    Input: tensor with shape [B, C, H, W]
    """
    def __init__(self, pool_size=3):
        super().__init__()
        self.pool = nn.AvgPool2d(
            pool_size, stride=1, padding=pool_size//2, count_include_pad=False)

    def forward(self, x):
        # Subtraction of the input itself is added
        # since the block already has a residual connection.
        return self.pool(x) - x
```

就是说，视觉 Transformer 的成功，不只是注意力机制的使用，Transformer 的整体架构（Norm、token mixer、残差，Norm Channel MLP、残差），对模型的效果提升有着更关键的作用，论文用一个 pooling 操作去替换注意力机制，就能取得很好的效果，论证了这一问题。

所以，我们在设计一个 xx-former 模型时，我们可以在 token mixer 部分，改进一个模块，这个算一个主要创新点，然后在 Channel MLP 部分改进一个模块，这算一个次创新点。